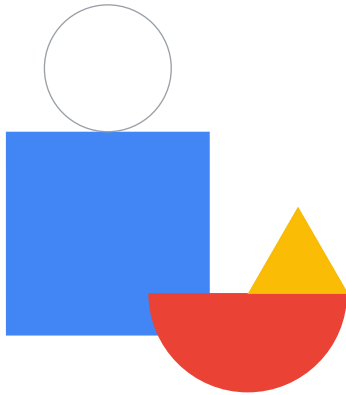


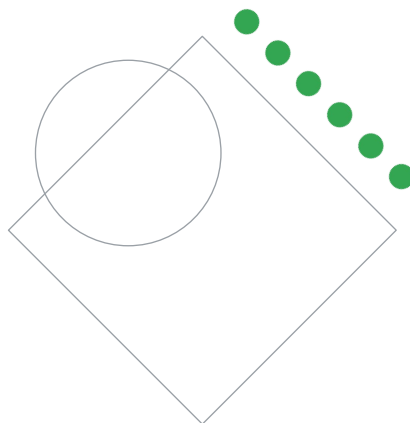


Preparing for Your Professional Cloud Network Engineer Journey

Module 6: Configure, manage and implement a cloud network security solution

Welcome to Module 6: Configure, manage and implement a cloud network security solution.

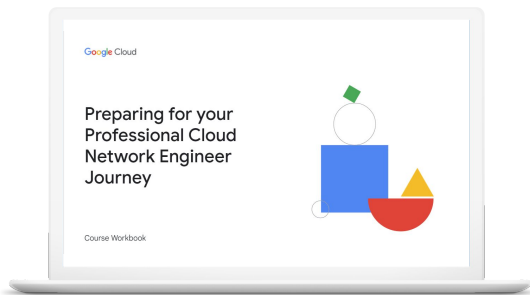
Review and study planning



You'll now review the diagnostic questions and your answers to help you identify what to include in your study plan.

Your study plan:

Configure, manage and implement a cloud network security solution



6.1

Configuring Google Cloud Armor policies

6.2

Configuring and managing Cloud Next Generation Firewall (NGFW) policies and VPC firewall rules

6.3

Configuring and securing internet egress traffic using Public Cloud NAT and Secure Web Proxy

6.4

Configuring self-managed network packet inspection, Cloud IDS, and Packet Mirroring

Google Cloud

The diagnostic questions align with these objectives of this exam section. Use the PDF resource that follows to review the questions and how you answered them. Pay specific attention to the rationale for both the correct and incorrect answers. Use the resources detailed under **Where to look** and **Content mapping** to build a study plan that meets your learning needs.

6.1 | Configuring Google Cloud Armor policies

Considerations include:

- Configuring and attaching edge and Backend security policies.
- Implementing web application firewall (WAF) rules (e.g., SQL injection, cross-site scripting, remote file inclusion).
- Configuring advanced network DDoS and Adaptive Protection.
- Configuring rate limiting.
- Configuring bot management.
- Applying Google Threat Intelligence.

Google Cloud

A Professional Cloud Network Engineer should be familiar with tools available to secure workloads against DDoS and other common attacks. Web application firewalls (WAF) are a popular class of tool for that purpose. In Google Cloud, Google Cloud Armor provides a managed WAF service.

You encountered these types of considerations in questions 1 and 2.

To answer these questions, you should be comfortable with all of the considerations listed on this slide.

6.1 Diagnostic Question 01 Discussion



Cymbal Bank wants to protect their services, which are deployed behind an Application Load Balancer, from L7 distributed denial of service (DDoS), SQL injection (SQLi), and cross-site scripting (XSS) attacks.

- A. **Configure Google Cloud Armor with the appropriate rules.**
- B. Configure a VM with appropriate scanning and filtering software in front of the Application Load Balancer.
- C. Configure Google Cloud WAF with the appropriate rules.
- D. Configure Cloud NAT with the appropriate rules.

Select the simplest approach to accomplish this.

Google Cloud

Feedback:

- *A. Correct! Google Cloud Armor can be easily configured to protect against these types of attacks.
- B. Incorrect. This can accomplish the desired effect but would not be as simple, scalable or cost effective as using Google Cloud Armor for that purpose.
- C. Incorrect. Google Cloud Armor is the managed WAF service for Google Cloud, and there is no product named Google Cloud WAF.
- D. Incorrect. Cloud NAT would not be able to satisfy the specified requirements

Where to look:

<https://cloud.google.com/armor/docs/security-policy-overview>
<https://cloud.google.com/armor/docs/configure-security-policies>
<https://cloud.google.com/armor/docs/rules-language-reference>
<https://cloud.google.com/armor/docs/rule-tuning>
<https://cloud.google.com/armor/docs/cloud-armor-overview>

Content mapping:

- ILT course: **Networking in Google Cloud**
 - M11 Hybrid Load Balancing and Traffic Management
 - M12 Caching and Optimizing Load Balancing
- On-demand course: **Networking in Google Cloud: Load Balancing**
 - M1 Hybrid Load Balancing and Traffic Management
 - M2 Caching and Optimizing Load Balancing

- Skill badges:
 - Build a Secure Google Cloud Network
 - Automate Deployment and Manage Traffic on a Google Cloud Network

Summary:

Google Cloud Armor can provide WAF capabilities via an expression language that allows logical combinations of various clauses in allow or deny rules. Allow rules allow matching traffic and block all other traffic, while deny rules block matching traffic and allow all other traffic. Allow and deny rules can be combined in a priority order.

Clauses can incorporate attack patterns or attributes of the request, such as source IP address, headers being present, or their values.

6.1 Diagnostic Question 02 Discussion



Cymbal Bank's public-facing login portal (HTTPS) is experiencing a brute-force attack. The attack traffic is coming from valid IP addresses but at an unusually high volume per user. You need to stop this specific attack vector without blocking legitimate low-volume traffic from the same regions. The solution must be implemented at the edge.

Select the simplest approach to accomplish this.

- A. Configure a VPC Firewall rule to deny ingress traffic on TCP port 443 from the suspect regions.
- B. Configure Google Cloud Armor with a rate-limiting rule to throttle requests based on the client IP address.**
- C. Configure Cloud NAT with a larger pool of IP addresses to handle the increased connection volume.
- D. Configure a Google Cloud Armor security policy with a preconfigured WAF rule for SQL injection.

Google Cloud

Feedback:

A. Incorrect. VPC firewall rules operate at Layer 3/4 and blocking a region affects all traffic from that location. This would block legitimate users alongside attackers, violating the requirement to maintain access.

*B. Correct! Google Cloud Armor rate limiting effectively mitigates brute-force attacks by throttling excessive requests per client IP.

C. Incorrect. Cloud NAT is designed for outbound (egress) connectivity for private VMs, not for securing inbound traffic to a Load Balancer. Increasing IP pool size addresses port exhaustion, not volumetric attacks.

D. Incorrect. SQL injection rules look for specific malicious code patterns in the payload, whereas brute-force attacks usually consist of valid strings or random credentials. A volume-based attack would likely bypass a signature-based filter.

Where to look:

<https://cloud.google.com/armor/docs/security-policy-overview>

<https://cloud.google.com/armor/docs/configure-security-policies>

<https://docs.cloud.google.com/armor/docs/rate-limiting-overview>

<https://cloud.google.com/armor/docs/cloud-armor-overview>

Content mapping:

- On-demand course: **Networking in Google Cloud: Network Security**
 - M1 Distributed Denial of Service (DDoS) Protection
- On-demand course: **Networking in Google Cloud: Load Balancing**

- M1 Hybrid Load Balancing and Traffic Management
- M2 Caching and Optimizing Load Balancing
- Skill badges:
 - Build a Secure Google Cloud Network
 - Automate Deployment and Manage Traffic on a Google Cloud Network
 - Designing Network Security in Google Cloud

Summary:

Google Cloud Armor allows you to configure rate-limiting rules. These rules define a maximum number of requests per client IP (or other keys) within a specified time interval. This prevents brute-force attacks by throttling or banning clients exceeding the threshold, whereas a VPC firewall would block all traffic (L3/L4) from the region, and SQL injection rules do not address volumetric brute-force attacks.

6.1 | Configuring Google Cloud Armor policies

Courses



[Networking in Google Cloud](#)

- M11 Hybrid Load Balancing and Traffic Management
- M12 Caching and Optimizing Load Balancing



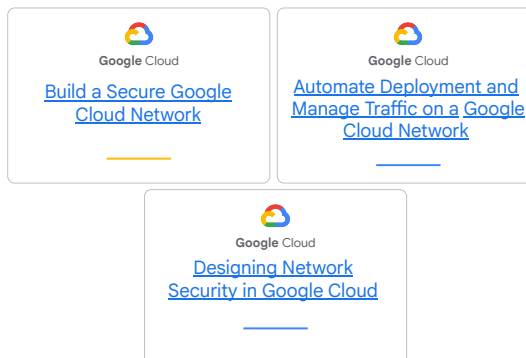
[Networking in Google Cloud: Load Balancing](#)

- M1 Hybrid Load Balancing and Traffic Management
- M2 Caching and Optimizing Load Balancing

[Networking in Google Cloud: Network Security](#)

- M1 Distributed Denial of Service (DDoS) Protection

Skill Badges



Documentation

[Security policy overview](#)
[Configuring Google Cloud Armor security policies](#)
[Rate limiting overview](#)
[Google Cloud Armor custom rules language reference](#)
[Tuning Google Cloud Armor WAF rules](#)
[Product overview](#)

Let's consider resources that can help you build your knowledge and skills in this area.

The concepts in the diagnostic question we just reviewed are covered in these modules and in this documentation. You'll find this list in your workbook so you can take a note of what you want to include later when you build your study plan. Based on your experience with the diagnostic questions, you may want to include some or all of these.

[Networking in Google Cloud \(ILT\)](#)

[Networking in Google Cloud: Load Balancing \(On-demand\)](#)

[Build a Secure Google Cloud Network \(Skill badge\)](#)

[Automate Deployment and Manage Traffic on a Google Cloud Network \(Skill badge\)](#)

[Designing Network Security in Google Cloud \(Skill badge\)](#)

<https://cloud.google.com/armor/docs/security-policy-overview>

<https://cloud.google.com/armor/docs/configure-security-policies>

<https://docs.cloud.google.com/armor/docs/rate-limiting-overview>

<https://cloud.google.com/armor/docs/rules-language-reference>

<https://cloud.google.com/armor/docs/rule-tuning>

<https://cloud.google.com/armor/docs/cloud-armor-overview>

6.2 | Configuring and managing Cloud Next Generation Firewall (NGFW) policies and VPC firewall rules

Considerations include:

- Planning the firewall strategy (e.g., VPC firewall rules, Cloud Next Generation Firewall, hierarchical firewall rules, or 3rd Party Integration).
- Configuring Cloud NGFW to support GKE.
- Creating and troubleshooting VPC firewall rules and Cloud NGFW regional/global/hierarchical policies.
- Enabling Layer 7 packet inspection with Cloud NGFW Enterprise.
- Migrating from VPC firewall rules to Cloud NGFW policies.
- Configuring VPC and NGFW firewall rule criteria (e.g. rule priority, network protocols, ingress and egress rules, health checks for backend services).

Google Cloud

The Professional Cloud Network Engineer uses firewall rules as the primary mechanism of traffic control with the VPCs and should be able to configure them to support all common workload communication scenarios. Considerations include target network tags and service accounts, rule priority, and network protocols.

You explored these considerations in the diagnostic questions. Question 3 tested your ability to configure firewall rules. Question 4 tested your ability to troubleshoot connectivity issues using firewall logging and insights. Question 5 explored your understanding of hierarchical firewall policies, and how to implement them with the least amount of administrative effort.

To answer these questions, you should be comfortable with all of the considerations listed on this slide and the next.

6.2 | Configuring and managing Cloud Next Generation Firewall (NGFW) policies and VPC firewall rules

Considerations include:

- Configuring VPC and NGFW Firewall Rules Logging.
- Incorporating micro segmentation for security purposes (e.g., using metadata, (secure) tags, service accounts, secure network tags).
- Differentiating between the different tiers of Cloud Next Generation Firewall: Essentials, Standard and Enterprise.

6.2 Diagnostic Question 03 Discussion



You are configuring firewall rules for securing a set of microservices (MS1, MS2, MS3) running in separate managed instance groups (MIGs) of VMs in a single subnet of a VPC network. The primary range of the VPC network is 10.128.128.0/20. MS1 will send requests to MS2 on TCP port 8443; MS2 will send requests to MS3 on TCP port 8663; and MS3 will send requests to MS1 on TCP port 8883. There will be no other communication to or between these microservices.

Select a simple and secure firewall configuration to support this traffic requirement.

- A. Create service accounts (S1, S2, S3) for the microservices, and assign those service accounts to the instance template for the MIG used by each microservice. Create three ingress allow firewall rules: for TCP 8443 from source S1 to target S2; for TCP 8663 from source S2 to target S3, and for TCP 8883 from source S3 to target S1.
- B. Create network tags (T1, T2, T3) for the microservices, and assign those network tags to the instance template for the MIG used by each microservice. Create three ingress allow firewall rules: for TCP 8443 from source T1 to target T2; for TCP 8663 from source T2 to target T3; and for TCP 8883 from source T3 to target T4.
- C. Create service accounts (S1, S2, S3) for the microservices, and assign those service accounts to the instance template for the MIG used by each microservice. Create three ingress allow firewall rules: for TCP 8443 from source 10.128.128.0/20 to target S2; for TCP 8663 from source 10.128.128.0/20 to target S3; for TCP and 8883 from source 10.128.128.0/20 to target S1'.
- D. Create network tags (T1, T2, T3) for the microservices, and assign those network tags to the instance template for the MIG used by each microservice. Create three ingress allow firewall rules: for TCP 8443 from source 10.128.128.0/20 to target T2; for TCP 8663 from source 10.128.128.0/20 to target T3; and for TCP 8883 from source 10.128.128.0/20 to target T1.

Feedback:

*A. Correct! This option is as simple as the others but provides better security: service accounts have tighter access control than network tags.

B. Incorrect. This option is slightly less secure than option A: service accounts have tighter access control than network tags.

C. Incorrect. This option is significantly less secure than option A. It would allow requests into the microservices from any other workloads deployed in the same subnet, using the same ports as the intended microservices.

D. Incorrect. This is significantly less secure than option A because it would allow requests into the microservices from any other workloads deployed in the same subnet using the same ports as the intended microservices.

Where to look:

<https://cloud.google.com/vpc/docs/firewalls>

<https://cloud.google.com/vpc/docs/using-firewalls>

Content mapping:

- ILT course: **Networking in Google Cloud**
 - M9 Controlling Access to VPC Networks
 - M11 Hybrid Load Balancing and Traffic Management
- On-demand course: **Networking in Google Cloud: Network Security**
 - M2 Controlling Access to VPC Networks

- On-demand course: **Networking in Google Cloud: Load Balancing**
 - M1 Hybrid Load Balancing and Traffic Management
- Skill badges:
 - Build a Secure Google Cloud Network
 - Implement Cloud Security Fundamentals on Google Cloud

Summary:

Firewall rules can be assigned to operate on all instances in a VPC. You can also assign them to specific VMs using source or destination IP addresses or IP ranges. You can also assign firewall rules using source and target service accounts or tags. In general using source and target service accounts is the recommended approach and provides the best security.

6.2 Diagnostic Question 04 Discussion



You are trying to determine which Next Generation Firewall (NGFW) rules are incorrectly blocking requests between two VMs running within a VPC network: VM1 and VM2. Firewall logging is enabled for all firewall rules, including metadata. The Firewall Insights and Recommendations API are also enabled. All insights are enabled, and an observation period is set over a period capturing the blocked requests.

Select a valid troubleshooting approach to find the incorrectly configured Next Generation Firewall (NGFW) rule.

- A. On the Firewall Insights page of the Google Cloud console, find the names of the deny firewall rules with hits to identify rules that are blocking requests. On the Legacy Logs Viewer or Logs Explorer page, view the firewall logs and filter for logs that match those rules by name, using `jsonPayload.rule_details.reference` field, matching the names of the deny firewall rules with hits.
- B. On the Logs Explorer page, view the firewall logs, and filter for logs that match the source and destination VMs VM1 and VM2, using the `jsonPayload.instance.project_id`, `jsonPayload.instance.vm_name`, `jsonPayload.instance.region`, and `jsonPayload.instance.zone`, `jsonPayload.remote_instance.vm_name`, `jsonPayload.remote_instance.region`, and `jsonPayload.remote_instance.zone`.
- C. On the Logs Explorer or Legacy Logs Viewer page, view the firewall logs, and filter for logs that match the destination VM2 in the VPC, using the `jsonPayload.instance.project_id`, `jsonPayload.instance.vm_name`, `jsonPayload.instance.region`, and `jsonPayload.instance.zone` fields.
- D. On the Firewall Insights landing page of the Google Cloud console, find the names of the allow firewall rules with no hits to identify rules that are not allowing requests. On the Logs Viewer or Explorer page, view the firewall logs and filter for logs matching those rules by name, using `jsonPayload.rule_details.reference` field (matching the names of the allow firewall rules with no hits).

Feedback:

A. Incorrect. This approach will not work if the traffic is being blocked by the implicit deny all ingress rule (i.e., there is no appropriate allow rule that is allowing the requests to ingress VM2).

*B. Correct! This is the only approach among the options that will detect the incorrectly configured rule whether it is a missing or incorrect ingress allow or a incorrect egress deny rule.

C. Incorrect. This approach will not work if the traffic is being blocked by some deny egress rule at VM1.

D. Incorrect. This approach will not work if the traffic is being blocked by some deny egress rule at VM1.

Where to look:

<https://cloud.google.com/vpc/docs/firewall-rules-logging>

<https://cloud.google.com/vpc/docs/using-firewall-rules-logging>

<https://cloud.google.com/network-intelligence-center/docs/firewall-insights/how-to/using-firewall-insights>

<https://cloud.google.com/network-intelligence-center/docs/firewall-insights/concepts/overview>

Content mapping (Partial coverage in Networking in Google Cloud):

- ILT course: **Networking in Google Cloud**
 - M3 Network Monitoring and Logging

- On-demand course: **Networking in Google Cloud: Fundamentals**
 - M3 Network Monitoring and Logging

Summary:

Firewall Logging and Firewall Insights can be useful tools for troubleshooting firewall configuration problems. They must be enabled and properly configured before usage. Firewall Insights provides high-level details for quick analysis and detection of problems, such as shadowed rules, overly permissive rules, and active deny rules. It doesn't directly provide detailed information about the firewall rule activity.

Firewall Insights can be combined with filtering on matching firewall logs to get more of the details. Firewall logs capture all of the detailed information about firewall activity but can produce large logs. These logs may require filtering to effectively find activity related to the traffic flows of interest.

When troubleshooting, it's also important to remember that logs and insights are not captured for the implicit deny all ingress and allow all egress firewall rules.

6.2 Diagnostic Question 05 Discussion



The Cymbal Bank security team requires that SSH access (TCP port 22) be blocked from the public internet for *all* projects within the "Banking-Prod" folder. Project administrators inside this folder must not be able to override this rule with their own VPC firewall rules.

- A. Create a VPC firewall rule in every VPC network within the folder to deny ingress on port 22.
- B. Create a hierarchical firewall policy at the Organization level with a priority of 0 to deny ingress on port 22.
- C. Create a hierarchical firewall policy applied to the "Banking-Prod" folder with a deny rule for ingress on port 22, using the `goto_next` action.
- D. **Create a hierarchical firewall policy applied to the "Banking-Prod" folder with a deny rule for ingress on port 22, and ensure it has a higher priority (lower number) than local VPC rules.**

Select the configuration which fulfills this requirement, but is implemented using the least administrative effort.

Feedback:

- A. Incorrect. Managing individual rules in every project VPC is operationally inefficient and prone to human error. Furthermore, project editors could disable these rules, failing the governance requirement.
- B. Incorrect. Applying the policy at the Organization level is too broad and would block SSH for the entire company. This violates the principle of least privilege, as the requirement was specific only to the "Banking-Prod" folder.
- C. Incorrect. The `goto_next` action explicitly delegates the decision to the next level (the local VPC rules). This would allow project administrators to override the folder-level policy and allow the traffic.
- *D. Correct! Hierarchical firewall policies take precedence over VPC rules, so a folder-level deny rule enforces a strict block that cannot be overridden.

Where to look:

<https://cloud.google.com/vpc/docs/firewall-rules-logging>
<https://cloud.google.com/vpc/docs/using-firewall-rules-logging>
https://docs.cloud.google.com/firewall/docs/using-firewall-policies#create_a_rule_in_a_n_existing_firewall_policy
<https://cloud.google.com/network-intelligence-center/docs/firewall-insights/concepts/overview>

Content mapping (Partial coverage in Networking in Google Cloud):

- On-demand course: **Networking in Google Cloud: Network Security**
 - M2 Controlling Access to VPC Networks

- Skill badge:
 - Designing Network Security in Google Cloud

Summary:

Hierarchical firewall policies allow you to define security rules at the organization or folder level. To enforce a block that cannot be overridden by local project rules, you must apply a deny rule at the folder level. Rules in hierarchical policies are evaluated before VPC firewall rules; if a hierarchical rule denies traffic, evaluation stops, and the traffic is dropped. `goto_next`` (Option C) would pass control down to lower layers, potentially allowing a local allow rule to accept the traffic.

6.2

Configuring and managing Cloud Next Generation Firewall (NGFW) policies and VPC firewall rules

Courses



[Networking in Google Cloud](#)

- M3 Network Monitoring and Logging
- M9 Controlling Access to VPC Networks
- M11 Hybrid Load Balancing and Traffic Management



[Networking in Google Cloud: Fundamentals](#)

- M3 Network Monitoring and Logging
- [Networking in Google Cloud: Network Security](#)
- M2 Controlling Access to VPC Networks
- [Networking in Google Cloud: Load Balancing](#)
- M1 Hybrid Load Balancing and Traffic Management

Skill Badges



Google Cloud

[Build a Secure Google Cloud Network](#)


Google Cloud

[Designing Network Security in Google Cloud](#)


Google Cloud

[Implement Cloud Security Fundamentals on Google Cloud](#)

Documentation

[VPC firewall rules overview](#)
[Using firewall rules | VPC](#)
[Firewall Rules Logging overview | VPC](#)
[Using Firewall Rules Logging | VPC](#)
[Use hierarchical firewall policies and rules](#)
[Using Firewall Insights](#)
[Firewall Insights overview](#)

Let's consider resources that can help you build your knowledge and skills in this area.

The concepts in the diagnostic questions we just reviewed are covered in these modules and in this documentation. You'll find this list in your workbook so you can take a note of what you want to include later when you build your study plan. Based on your experience with the diagnostic questions, you may want to include some or all of these.

[Networking in Google Cloud \(ILT\)](#)

[Networking in Google Cloud: Fundamentals \(On-demand\)](#)

[Networking in Google Cloud: Network Security \(On-demand\)](#)

[Networking in Google Cloud: Load Balancing \(On-demand\)](#)

[Build a Secure Google Cloud Network \(Skill badge\)](#)

[Implement Cloud Security Fundamentals on Google Cloud \(Skill badge\)](#)

[Designing Network Security in Google Cloud \(Skill badge\)](#)

<https://cloud.google.com/vpc/docs/firewalls>

<https://cloud.google.com/vpc/docs/using-firewalls>

<https://cloud.google.com/vpc/docs/firewall-rules-logging>

<https://cloud.google.com/vpc/docs/using-firewall-rules-logging>

https://docs.cloud.google.com/firewall/docs/using-firewall-policies#create_a_rule_in_a_n_existing_firewall_policy

<https://cloud.google.com/network-intelligence-center/docs/firewall-insights/how-to/using-firewall-insights>

<https://cloud.google.com/network-intelligence-center/docs/firewall-insights/concepts/overview>

6.3 | Configuring and securing internet egress traffic using Public Cloud NAT and Secure Web Proxy

Considerations include:

- Configuring public Cloud NAT IP addressing, assigning automatic and manual NAT IP addresses.
- Configuring static and dynamic port allocation for Cloud NAT.
- Configuring Secure Web Proxy.

The Professional Cloud Network Engineer understands that security must be balanced with workload requirements. In many cases, although access to private resources from the internet should be prohibited, access in the opposite direction will be required. Network address translation is a valuable tool to facilitate controlled access. Cloud NAT is a managed service that makes NAT in Google Cloud simple.

You explored these types of considerations for configuring Cloud NAT in question 6. Question 7 tested your knowledge of Secure Web Proxy and how to filter traffic.

6.3 Diagnostic Question 06 Discussion



Cymbal is using Cloud NAT to provide internet connectivity to a group of VMs in a subnet. There are 500 VMs in the subnet, and each VM may have up to 1000 internet bound connections simultaneously.

What Cloud NAT configuration will support this requirement?

- A. Set the minimum ports per VM to 1000, and set the number of IP addresses used by the Cloud NAT Gateway to 8.
- B. Set the minimum ports per VM to 2000, and set the number of IP addresses used by the Cloud NAT Gateway to 8.
- C. Set the minimum ports per VM to 2000, and set the number of IP addresses used by the Cloud NAT Gateway to 10.
- D. Set the minimum ports per VM to 1000, and set the number of IP addresses used by the Cloud NAT Gateway to 6.

Google Cloud

Feedback:

*A. Correct! 1000 ports per VM will be sufficient to provide up to 1000 internet-bound UDP and TCP connections to any destination 3-tuple (combination of destination IP, destination port, and protocol) per VM. With a minimum 1000 ports allocated per VM, each NAT IP address can support up to 64.5 VMs (64,512 ports per NAT IP address/1000 ports per VM), and 8 NAT IP addresses would be sufficient for 500 VMs.

B. Incorrect. 2000 ports per VM would be more than you need. With that setting, a single NAT IP address could only support 32.26 VMs, and 8 NAT IP addresses would only support 258 VMs.

C. Incorrect. 2000 ports per VM would be more than you need. With that setting, a single NAT IP address could only support 32.26 VMs, and 10 NAT IP addresses would only support 322 VMs.

D. Incorrect. 6 NAT IP addresses with 1000 ports per VM would only support 387 VMs.

Where to look:

<https://cloud.google.com/nat/docs/overview>
<https://cloud.google.com/nat/docs/ports-and-addresses>
<https://cloud.google.com/nat/docs/using-nat>
<https://cloud.google.com/nat/docs/gce-example>
<https://cloud.google.com/nat/docs/using-nat-rules>

Content mapping:

- ILT course: **Networking in Google Cloud**
 - M5 Private Connection Options
- On-demand course: **Networking in Google Cloud: Routing and Addressing**
 - M2 Private Connection Options

Summary:

Cloud NAT allows outbound internet requests for VMs without external IP addresses and will allow responses to those outbound requests to be forwarded back to the initiating VM. Cloud NAT can either automatically allocate the set of external IP addresses it uses, or they can be manually specified. Each NAT IP address can support 64512 TCP and 64512 UDP connections. The connections can be distributed among all the VMs using that NAT gateway based on the minimum ports per VM configuration. This minimum ports per VM value can be either statically or dynamically configured, but will create a limit on how many VMs can be served by each NAT IP address.

6.3 Diagnostic Question 07 Discussion



Cymbal Bank has a fleet of private GKE clusters that need to download software updates from a specific external vendor URL (`update.vendor-a.com`). Corporate security policy strictly prohibits these VMs from accessing any other website on the internet. Cloud NAT is already configured for internet connectivity.

What configuration would allow you to secure this egress traffic?

- A. **Deploy Secure Web Proxy and configure a rule to allow access to the FQDN `update.vendor-a.com` and deny all other traffic. Point the VMs to use the proxy.**
- B. Configure a VPC Firewall egress rule to allow traffic only to the resolved IP addresses of `update.vendor-a.com`.
- C. Update the Cloud NAT configuration to use manual IP allocation and allowlist those IPs at the vendor side.
- D. Use Google Cloud Armor to allowlist the egress traffic based on the destination hostname.

Google Cloud

Feedback:

- *A. Correct! Secure Web Proxy allows you to filter egress traffic based on Fully Qualified Domain Names (FQDNs) and URLs, which is required when destination IP addresses may change or are shared Content Delivery Networks (CDNs).
- B. Incorrect. Firewall rules rely on static IP ranges, but vendor update URLs often resolve to constantly changing Content Delivery Network (CDN) IPs. This makes maintaining an IP-based allowlist impossible.
- C. Incorrect. Cloud NAT provides connectivity but lacks the capability to filter traffic based on destination URLs or FQDNs. Allowlisting source IPs at the vendor does not prevent the VMs from accessing other unauthorized sites.
- D. Incorrect. Google Cloud Armor is an ingress security service for protecting applications from external attacks. It cannot be used to filter outbound (egress) traffic originating from internal VMs.

Where to look:

<https://cloud.google.com/nat/docs/overview>
<https://cloud.google.com/nat/docs/ports-and-addresses>
<https://cloud.google.com/nat/docs/using-nat>
<https://cloud.google.com/nat/docs/gce-example>
<https://cloud.google.com/nat/docs/using-nat-rules>

Content mapping:

- ILT course: **Networking in Google Cloud**
 - M9 Controlling Access to VPC Networks

- On-demand course: **Networking in Google Cloud: Network Security**
 - M2 Controlling Access to VPC Networks
- Skill badge:
 - Designing Network Security in Google Cloud

Summary:

Secure Web Proxy allows you to filter egress traffic based on Fully Qualified Domain Names (FQDNs) and URLs, which is required when destination IP addresses may change or are shared (CDNs). Cloud NAT (Option C) provides connectivity but does not filter based on URLs. VPC firewall rules (Option B) operate on IP ranges, making it difficult to manage FQDN-based filtering. Cloud Armor (Option D) is primarily for Ingress filtering, not Egress.

6.3 Configuring and securing internet egress traffic using Public Cloud NAT and Secure Web Proxy

Courses



[Networking in Google Cloud](#)

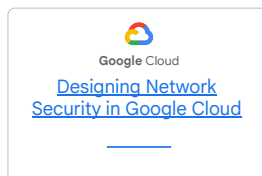
- M5 Private Connection Options
- M9 Controlling Access to VPC Networks



[Networking in Google Cloud: Routing and Addressing](#)

- M2 Private Connection Options
- [Networking in Google Cloud: Network Security](#)
- M2 Controlling Access to VPC Networks

Skill Badges



Documentation

- [Cloud NAT overview](#)
- [Cloud NAT address and port overview](#)
- [Configure Cloud NAT](#)
- [Example Compute Engine setup | Cloud NAT](#)
- [Using Cloud NAT rules](#)

Let's consider resources that can help you build your knowledge and skills in this area.

The concepts in the diagnostic questions we just reviewed are covered in these modules and in this documentation. It's highly recommended that you review the documentation. You'll find this list in your workbook so you can take a note of what you want to include later when you build your study plan. Based on your experience with the diagnostic questions, you may want to include some or all of these.

[Networking in Google Cloud \(ILT\)](#)

[Networking in Google Cloud: Routing and Addressing \(On-demand\)](#)

[Networking in Google Cloud: Network Security \(On-demand\)](#)

[Designing Network Security in Google Cloud \(Skill badge\)](#)

<https://cloud.google.com/nat/docs/overview>

<https://cloud.google.com/nat/docs/ports-and-addresses>

<https://cloud.google.com/nat/docs/using-nat>

<https://cloud.google.com/nat/docs/gce-example>

<https://cloud.google.com/nat/docs/using-nat-rules>

6.4 | Configuring self-managed network packet inspection, Cloud IDS, and Packet Mirroring

Considerations include:

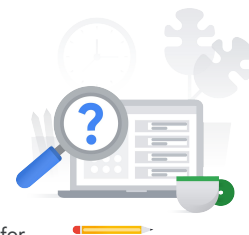
- Routing and inspecting inter-VPC traffic using multi-NIC VMs (e.g., next-generation firewall appliances).
- Configuring an internal load balancer as a next hop for highly available multi-NIC VM routing.
- Configure Policy Based Routes for highly available multi-NIC VM routing.
- Developing a strategy for Out-of-band Network Security Integration (NSI).
- Configuring the Cloud Intrusion Detection System (Cloud IDS).
- Configuring Packet Mirroring for VPC traffic towards self-managed collectors.

Google Cloud

Professional Cloud Network Engineers should be familiar with tools and processes for efficient collection and analysis of packet captures for real-time or batch security scanning and filtering, forensics, or performance analysis. In Google Cloud, two primary tools are VMs with multiple network interfaces (multi-NIC) and Packet Mirroring.

Question 8 asked you to apply scenarios using VMs with Multi-NIC. Question 9 tested your knowledge of concepts related to packet mirroring. Question 10 tested your ability to configure Cloud IDS to detect malware, spyware and attacks.

6.4 Diagnostic Question 08 Discussion



You are designing a system in Google Cloud to ensure that all traffic being sent between two subnets is passed through a security gateway VM. The VM runs third-party software that scans traffic for known attack signatures and then forwards or drops traffic based on the scan results.

Select a configuration that satisfies these requirements.

- A. Create the two subnets in the same VPC. Create a VM running the third-party scanning software in one of the subnets. Create custom routes in the VPC to send traffic for each subnet from the opposite subnet through that VM.
- B. Create the two subnets in the same VPC. Create a VM running the third-party scanning software in each of the subnets. Create custom routes in the VPC to send traffic destined for each subnet originating in the opposite subnet through the VM in its subnet.
- C. **Create the two subnets in two separate VPCs. Create a VM with two network interfaces (NICs), with each NIC connected to the subnet in each VPC. Create custom routes in each VPC to send traffic destined for each subnet originating in the opposite subnet through the VM.**
- D. Create the two subnets in the same VPC. Create two VMs running the third-party scanning software, with one in each of the subnets. Create custom routes in the VPC to send traffic destined for each subnet originating in the opposite subnet through the VM in the opposite subnet.

Google Cloud

Feedback:

A. Incorrect. This configuration is not possible. Custom routes cannot match or be contained in subnet routes, so there would be no way to enforce subnet bound traffic to go through the VM.

B. Incorrect. This configuration is not possible. Custom routes cannot match or be contained in subnet routes, so there would be no way to enforce subnet bound traffic to go through the VM.

*C. Correct! This configuration is the only option that can accomplish the requirements. Custom routes can be created in each VPC that match the subnet range of the opposite subnet with a next hop of the gateway VM. The gateway VM can then forward or drop the traffic based on scan results.

D. Incorrect. This configuration is not possible. Custom routes cannot match or be contained in subnet routes, so there would be no way to enforce subnet bound traffic to go through the VM.

Where to look:

<https://cloud.google.com/vpc/docs/multiple-interfaces-concepts>

<https://cloud.google.com/vpc/docs/create-use-multiple-interfaces>

<https://cloud.google.com/load-balancing/docs/internal/ilb-next-hop-overview>

<https://cloud.google.com/load-balancing/docs/internal/setting-up-ilb-next-hop>

Content mapping:

- ILT course: **Networking in Google Cloud**
 - M1 VPC Networking Fundamentals

- On-demand course: **Networking in Google Cloud: Fundamentals**
 - M1 VPC Networking Fundamentals

Summary:

VMs with multiple NICs can be used to connect multiple VPCs. Each NIC in a VM with multiple NICs must be in a separate VPC and each VPC can have its own independent set of custom routes (in addition to the default routes) and custom routes can have next hops on a VM with multiple NICs that act as a gateway between the VPCs. This pattern can be useful in scenarios where VPCs with overlapping subnet ranges need to be connected for private IP communication, or in cases where traffic needs to be scanned between trusted and untrusted networks without using public IP addresses.

6.4 Diagnostic Question 09 Discussion



Select the list of the resources that must be created or configured to enable Packet Mirroring.

- A. A Packet Mirroring policy and a collector instance
- B. A pPacket Mirroring policy, an internal passthrough Network Load Balancer configured for Packet Mirroring, an instance group of collector instances, and firewall rules**
- C. A Packet Mirroring policy, a collector instance, and firewall rules
- D. A Packet Mirroring policy, an instance group of collector instances, and firewall rules

Google Cloud

Feedback:

A. Incorrect. This list is missing necessary components: an internal passthrough Network Load Balancer configured for Packet Mirroring, an instance group of collector instances, and firewall rules.

*B. Correct! This list contains the necessary component resources.

C. Incorrect. This list is missing necessary components: an internal passthrough Network Load Balancer configured for Packet Mirroring and an instance group of collector instances.

D. Incorrect. This list is missing a necessary component: an internal passthrough Network Load Balancer configured for Packet Mirroring.

Where to look:

<https://cloud.google.com/vpc/docs/packet-mirroring>

<https://cloud.google.com/vpc/docs/using-packet-mirroring>

<https://cloud.google.com/vpc/docs/monitoring-packet-mirroring>

<https://cloud.google.com/load-balancing/docs/internal>

Content mapping:

- ILT course: **Networking in Google Cloud**
 - M3 Network Monitoring and Logging
 - M10 Advanced Security Monitoring and Analysis
- On-demand course: **Networking in Google Cloud: Fundamentals**
 - M3 Network Monitoring and Logging

- On-demand course: **Networking in Google Cloud: Network Security**
 - M3 Advanced Security Monitoring and Analysis
- Skill badge: Implement Cloud Security Fundamentals on Google Cloud

Summary:

Each Packet Mirroring policy sends traffic from a collection of source VMs, which must all be in the same project, VPC, and region, to a destination internal TCP/UDP load balancer configured for Packet Mirroring. The source VMs can be specified in the Packet Mirroring policy by name, tag, or subnet, with different limitations depending on the approach taken. The destination load balancer can be connected to one or more VMs in an unmanaged or managed instance group. It must also be in the same region, but can be in either the same VPC or a peered VPC. Packet Mirroring policies can be configured to forward only ingress or only egress traffic, or to limit the traffic to only include certain source or destination IP ranges and/or protocols.

6.4 Diagnostic Question 10 Discussion



Cymbal Bank requires deep packet inspection for traffic moving East-West between two VPCs to detect malware and command-and-control attacks. The team wants a managed solution that provides threat detection signatures automatically without the operational overhead of managing collector VMs or load balancers.

What solution would you deploy to meet the criteria?

- A. Configure VPC Flow Logs with metadata enabled, and stream logs to BigQuery for analysis.
- B. Configure Packet Mirroring to forward traffic to a Managed Instance Group of 3rd-party NGFW appliances.
- C. Deploy Google Cloud IDS (Intrusion Detection System) endpoints in the VPCs.**
- D. Implement Google Cloud Armor with Adaptive Protection on the internal load balancers.

Google Cloud

Feedback:

A. Incorrect. VPC Flow Logs only capture connection metadata (who talked to whom), not the packet payload. They cannot detect malware signatures or command-and-control patterns within the data.

B. Incorrect. This approach requires you to manage, scale, and update the third-party collector VMs yourself. The scenario explicitly requested a managed solution with "no operational overhead."

*C. Correct! Cloud IDS is a managed service that detects malware, spyware, and command-and-control attacks. It fits the requirement of "managed solution" and "no operational overhead of managing collector VMs."

D. Incorrect. Google Cloud Armor is a WAF designed to protect Load Balancers from L7 web attacks. It is not designed to inspect internal East-West traffic between VPCs for general malware.

Where to look:

<https://cloud.google.com/vpc/docs/packet-mirroring>

<https://cloud.google.com/vpc/docs/using-packet-mirroring>

<https://cloud.google.com/vpc/docs/monitoring-packet-mirroring>

<https://cloud.google.com/load-balancing/docs/internal>

<https://docs.cloud.google.com/intrusion-detection-system/docs/configuring-ids>

Content mapping:

- ILT course: **Networking in Google Cloud**

- M9 Controlling Access to VPC Networks
- On-demand course: **Networking in Google Cloud: Network Security**
 - M2 Controlling Access to VPC Networks
- Skill badge:
 - Designing Network Security in Google Cloud

Summary:

Cloud IDS is a managed service that detects malware, spyware, and command-and-control attacks. It fits the requirement of "managed solution" and "no operational overhead of managing collector VMs." Packet Mirroring (Option B) requires managing collector instances (DIY). VPC Flow Logs (Option A) only provide metadata (who talked to whom), not deep packet inspection (payload). Cloud Armor (Option D) is a WAF, primarily for L7 web attacks, not general malware detection in east-west traffic.

6.4 | Configuring self-managed network packet inspection, Cloud IDS, and Packet Mirroring

Courses



[Networking in Google Cloud](#)

- M1 VPC Networking Fundamentals
- M3 Network Monitoring and Logging
- M9 Controlling Access to VPC Networks
- M10 Advanced Security Monitoring and Analysis



[Networking in Google Cloud: Fundamentals](#)

- M1 VPC Networking Fundamentals
- M3 Network Monitoring and Logging

[Networking in Google Cloud: Network Security](#)

- M2 Controlling Access to VPC Networks
- M3 Advanced Security Monitoring and Analysis

Skill Badges



Google Cloud

[Implement Cloud Security Fundamentals on Google Cloud](#)



Google Cloud

[Designing Network Security in Google Cloud](#)

Documentation

[Multiple network interfaces overview and examples | VPC](#)

[Creating instances with multiple network interfaces | VPC](#)

[Internal TCP/UDP load balancers as next hops | Load Balancing](#)

[Setting up Internal TCP/UDP Load Balancing for third-party appliances](#)

[Packet Mirroring overview | VPC](#)

[Using Packet Mirroring | VPC](#)

[Monitoring Packet Mirroring | VPC](#)

[Internal passthrough Network Load Balancer overview](#)

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[Networking in Google Cloud \(ILT\)](#)

[Networking in Google Cloud: Fundamentals \(On-demand\)](#)

[Networking in Google Cloud: Network Security \(On-demand\)](#)

[Implement Cloud Security Fundamentals on Google Cloud \(Skill badge\)](#)

[Designing Network Security in Google Cloud \(Skill badge\)](#)

<https://cloud.google.com/vpc/docs/multiple-interfaces-concepts>

<https://cloud.google.com/vpc/docs/create-use-multiple-interfaces>

<https://cloud.google.com/load-balancing/docs/internal/ilb-next-hop-overview>

<https://cloud.google.com/load-balancing/docs/internal/setting-up-ilb-next-hop>

<https://cloud.google.com/vpc/docs/packet-mirroring>

<https://cloud.google.com/vpc/docs/using-packet-mirroring>

<https://cloud.google.com/vpc/docs/monitoring-packet-mirroring>